

Buzzard cluster mock data vector with optical-selection effects: forward model, catalog stacking, and analytical $\mathcal{B}_{\text{sel}}$ closure

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Abstract

This document is the technical reference for the `Xin.MockDataVector.ipynb` forward model: a Buzzard-halo DES Y3-like cluster mock that layers the Costanzi+2026 (C26) optical selection systematics on top of the raw simulation outputs to produce the two pieces a likelihood needs — a binned cluster number-count vector $N(\lambda^{\text{ob}}, z^{\text{ob}})$ and a binned tangential-shear data vector $\langle \gamma_t(R) | \lambda^{\text{ob}}, z^{\text{ob}} \rangle$. The pipeline draws an intrinsic richness λ^{tr} from the C26 mass–richness relation (Eq. 15), projects it to an observed richness λ^{ob} via the Costanzi+2019 (C19) Eq. 6 mixture (Gaussian core + EMG projection tail), and constructs the lensing data vector along two parallel routes: a catalog-driven empirical $\mathcal{B}_{\text{sel}}$ ratio that stacks `LAMBDA_CHISQ/Z_LAMBDA` selected clusters and divides by an (M, z_{true}) -matched random reference (Wu+2022, Costanzi+2026), and an analytical C26 Eq. C1 fitting function applied to the C19 forward-model stack. Boost-factor contamination is left for a follow-up. All code lives in this directory; the production notebook is `Xin.MockDataVector.ipynb` and the canonical output is `/global/u2/x/xintang/mocks/mock/Xin.MockDataVector.npz`.

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1 Pipeline topology

The notebook executes the following sequence of stages:

```
halo_load → photoz_proxy → sample_lambda_true → sample_lambda_obs → number_counts
→ stack_sigma → redmapper_bin → mass_match_random → B_sel_emp → C1_analytical →
save_npz
```

The first three stages are the forward model: select host halos above $\log_{10} M_{\text{vir}} = 13$ in $z \in [0.20, 0.65]$ (with the Buzzard $0.33 \leq z \leq 0.37$ box seam excised), add a DES Y3-like photo- z scatter to produce z^{ob} , and draw λ^{tr} and λ^{ob} from the C26/C19 kernels. The next four stages build the lensing data vector: stack $\Sigma/\Delta\Sigma$ in $(\lambda^{\text{ob}}, z^{\text{ob}})$ bins, re-stack on the catalog redMaPPer outputs (LAMBDA_CHISQ, Z_LAMBDA), construct the (M, z_{true}) -matched random reference, and form $\mathcal{B}_{\text{sel}}^{\text{emp}}(R)$. The last two stages emit the analytical C1 corrections that an MCMC chain would consume in place of the per-sample stack, and persist everything to a single `.npz` archive.

Table 1 summarises the role of each stage in the notebook.

Stage	Source	Role
<code>halo_load</code>	<code>fileLoc.py</code> + <code>fitsio</code>	raw Buzzard halos + per-halo $\Sigma, \Delta\Sigma$
<code>photoz_proxy</code>	inline notebook	Gaussian $z^{\text{ob}} = z^{\text{tr}} + \sigma_z(1 + z^{\text{tr}})\mathcal{N}(0, 1)$
<code>sample_lambda_true</code>	<code>costanzi_selection.py</code>	C26 Eq. 15 compound-Poisson + lognormal
<code>sample_lambda_obs</code>	<code>costanzi_selection.py</code>	C19 Eq. 6 mixture (Gaussian core + EMG tail)
<code>number_counts</code>	inline notebook	$N(\lambda^{\text{ob}}, z^{\text{ob}})$ binned counts
<code>stack_sigma</code>	inline notebook	$\langle \Sigma(R) \rangle, \langle \Delta\Sigma(R) \rangle$ in (λ, z) bins
<code>redmapper_bin</code>	inline notebook	re-stack on LAMBDA_CHISQ, Z_LAMBDA
<code>mass_match_random</code>	<code>stacked_profile_weighted_by_mass_redshift.py</code>	Wu+2022 (M, z^{tr}) -matched reference
<code>B_sel_emp</code>	inline notebook	ratio of selected to mass-matched stack
<code>C1_analytical</code>	inline notebook	C26 Appendix C Eq. C1 fit, Σ and $\Delta\Sigma$ flavours
<code>save_npz</code>	inline notebook	write <code>Xin_MockDataVector.npz</code>

Table 1: Notebook stage inventory. “Source” is the file the heavy lifting lives in; everything labelled “inline notebook” is direct numpy code in `Xin_MockDataVector.ipynb`.

2 Shared conventions

Cosmology. FlatLambdaCDM($H_0=70$, $\Omega_m=0.3$, $\Omega_b=0.046$, $T_{\text{cmb}}=2.725$) via `astropy.cosmology`, matching the Buzzard simulation cosmology used by the upstream halo catalogue (`cosmo0` in cell 3 of the notebook).

Units. Per-halo $\Sigma, \Delta\Sigma$ from `floc.profile_output.fname` are stored in physical $M_\odot/\text{pc}^2$ (FITS header TUNIT9, TUNIT10). Radii `rp_phys_mpc` (from `radial_bins_phys_mpc.py`) are in physical Mpc, with $N_R = 15$ log-spaced bins between 0.0323 and 30 Mpc. Halo masses `Mvir` are in $h^{-1} M_\odot$ (FITS header TUNIT2); the C26 mass–richness parameters are quoted in $M_\odot$, so `Mvir` is multiplied by h on read where the comparison demands it (e.g. the `hmf` cross-check in §7).

Halo selection. Mirrors `0-MakeMock.ipynb: pid == -1` (host halos only), $0 \leq \cos i \leq 1$ (measured), drop the simulation-box seam $0.33 \leq z < 0.37$, then require $0.20 \leq z^{\text{tr}} \leq 0.65$ and $\log_{10} M_{\text{vir}} \geq 13$. This yields $\sim 5.83 \times 10^5$ halos.

Binning. The default richness edges are `LBDBINS = {5, 15, 25, 40, 160}` (4 bins) and the photo- z edges are `ZMIN_LIST = {0.20, 0.37, 0.51}`, `ZMAX_LIST = {0.32, 0.51, 0.64}` (3 bins). These match the legacy `createDataVector.ipynb` edges so the data vector saved here can be cross-compared. An alternative DES Y3-style binning (`{20, 30, 45, 60, 500}`, `{0.20, 0.35, 0.50, 0.65}`) is also supported in the analytical C1 panel of §9.

Photo- z proxy. The DES Y3 redshift uncertainty is modelled by a per-halo Gaussian

$$z^{\text{ob}} = z^{\text{tr}} + \sigma_z(1 + z^{\text{tr}})\mathcal{N}(0, 1), \quad \sigma_z = 0.01, \quad (1)$$

clipped at $z^{\text{ob}} \geq 0.02$ to avoid unphysical negatives. This is a deliberate stand-in for the redshift- and colour-dependent kernel of Myles+2021; it captures the dominant scatter but not the tails. Throughout the notebook λ^{ob} is drawn at z^{ob} (not z^{tr}), and all observed-frame binning uses z^{ob} for self-consistency with the projection kernel.

Random seeds. All Monte-Carlo draws use a single `numpy.random.default_rng(seed=42)` shared across cells, except the analytical-PDF sanity histogram (§4) which uses `seed=123` and the $\mathcal{B}_{\text{sel}}$ bootstrap which uses `seed=0`.

3 The intrinsic mass–richness relation (C26 Eq. 15)

`Path` `costanzi.selection.py (l_sat, l_tr, sample_lambda_true)`. Implements the C26 forward model for the intrinsic richness λ^{tr} as a function of halo mass M and redshift z .

3.1 Mathematical form

The mean satellite richness follows C26 Eq. 15,

$$\langle \lambda_{\text{sat}} | M, z \rangle = \left(\frac{M - M_{\text{min}}}{M_{\text{pivot}}} \right)^\alpha \left(\frac{1 + z}{1 + z_{\text{piv}}} \right)^\epsilon, \quad (2)$$

clipped to zero below M_{min} to avoid a fractional power of a negative number. The mean intrinsic richness adds a central contribution that is unity above threshold and zero below (C26 Sec. III):

$$\langle \lambda^{\text{tr}} | M, z \rangle = \begin{cases} 1 + \langle \lambda_{\text{sat}} | M, z \rangle, & M \geq M_{\text{min}}, \\ 0, & M < M_{\text{min}}. \end{cases} \quad (3)$$

The DES Y1 NC+3×2pt best-fit constants hard-coded in `costanzi.selection.py` are

$$\log_{10} M_{\text{min}} = 11.385, \quad \alpha = 0.859, \quad \log_{10} M_1 = 12.696, \quad \sigma_{\text{intr}} = 0.181, \quad \epsilon = 0.284, \quad z_{\text{piv}} = 0.4544, \quad (4)$$

with $M_{\text{pivot}} = M_1 - M_{\text{min}}$.

The per-halo λ^{tr} is drawn from a compound Poisson with a multiplicative-lognormal broadening of the Poisson mean (C26 Sec. III, generative form):

$$m_{\text{broad}} = m_{\text{sat}} \exp\left[\sigma_{\text{intr}} \mathcal{N}(0, 1) - \frac{1}{2}\sigma_{\text{intr}}^2\right], \quad N_{\text{sat}} \sim \text{Poisson}(m_{\text{broad}}), \quad (5)$$

with the central galaxy contributing the explicit +1 (and only above M_{min}):

$$\lambda^{\text{tr}} = \mathbb{I}[M \geq M_{\text{min}}] + N_{\text{sat}}. \quad (6)$$

The lognormal mean is set so that $E[m_{\text{broad}}] = m_{\text{sat}}$ exactly, preserving the first moment of the Poisson–Gaussian convolution stated in C26 to second order in σ_{intr} .

3.2 Algorithm

Algorithm 1: `sample_lambda_true(M, z, rng)`

Knobs: `M_MIN`, `M_PIVOT`, `ALPHA`, `EPSI`, `PIVOT_Z0`, `SIGM_INTR`

Inputs: M (halo mass, $h^{-1} M_{\odot}$), z (halo redshift)

Outputs: λ^{tr} per halo, integer-valued

```

1  $m_{\text{sat}} \leftarrow \text{l\_sat}(M, z)$  via Eq. (2)
2  $m_{\text{broad}} \leftarrow m_{\text{sat}} \cdot \text{rng.lognormal}(\mu = -\frac{1}{2}\sigma_{\text{intr}}^2, \sigma = \sigma_{\text{intr}})$ 
3  $m_{\text{broad}} \leftarrow \text{clip}(m_{\text{broad}}, 0, \infty)$ 
4  $N_{\text{sat}} \leftarrow \text{rng.poisson}(m_{\text{broad}})$ 
5 return  $(M \geq M_{\text{min}}) + N_{\text{sat}}$ 

```

Sanity checks (notebook cells 11–12). The notebook overlays the population-mean curve $\langle \lambda^{\text{tr}} | M, z \rangle$ at three reference redshifts on a hexbin of the per-halo draws to confirm the generative mean matches the analytical mean curve, and histograms the residual $\Delta \log_{10} \lambda^{\text{tr}} = \log_{10} \lambda^{\text{tr}} - \log_{10} \langle \lambda^{\text{tr}} | M, z \rangle$ to expose the asymmetric scatter. The expected $\langle \lambda^{\text{tr}} \rangle$ on the post-selection mock is ~ 5.5 , with the fraction above $\lambda^{\text{tr}} = 20$ at the percent level — consistent with the low-mass-dominated halo catalogue.

4 The projection kernel (C19 Eq. 6)

`Path` `costanzi.selection.py` (`projection.params`, `sample_lambda_obs`). Maps $\lambda^{\text{tr}} \rightarrow \lambda^{\text{ob}}$ via the four-parameter Costanzi+2019 mixture model calibrated against SDSS redMaPPer.

4.1 Mathematical form

C19 Eq. 6 is a Gaussian-core + exponentially-modified-Gaussian (EMG) mixture:

$$P(\lambda^{\text{ob}} | \lambda^{\text{tr}}, z) = (1 - f_{\text{prj}}) \mathcal{N}(\mu, \sigma) + f_{\text{prj}} \text{EMG}(\mu, \sigma, \tau), \quad (7)$$

with the EMG component

$$\text{EMG}(\lambda^{\text{ob}}; \mu, \sigma, \tau) = \frac{\tau}{2} \exp\left[\frac{\tau}{2}(2\mu + \tau\sigma^2 - 2\lambda^{\text{ob}})\right] \text{erfc}\left(\frac{\mu + \tau\sigma^2 - \lambda^{\text{ob}}}{\sqrt{2}\sigma}\right). \quad (8)$$

The four parameters are linear in their per-redshift coefficients loaded from `prj_params_DESY3_lss_lin_dep_getdist_v1` (C19 Appendix A, Eq. A8):

$$\mu(\lambda^{\text{tr}}, z) = a_{\mu}(z) + b_{\mu}(z) \lambda^{\text{tr}}, \quad \sigma(\lambda^{\text{tr}}, z) = b_{\sigma}(z) (\lambda^{\text{tr}})^{a_{\sigma}(z)}, \quad (9)$$

$$\tau(\lambda^{\text{tr}}, z) = b_{\tau}(z) (\lambda^{\text{tr}})^{-a_{\tau}(z)}, \quad f_{\text{prj}}(\lambda^{\text{tr}}, z) = b_{f_{\text{prj}}}(z) (1 + e^{-\lambda^{\text{tr}}})^{-a_{f_{\text{prj}}}(z)}. \quad (10)$$

The 15 rows of the parameter file are interpreted here as posterior samples from the upstream get-dist chain (NOT redshift knots — the z -dependence is encoded in the linear coefficients themselves); `load_prj_posterior_mean` collapses them to the posterior mean, which is the operating point used throughout the notebook.

4.2 Algorithm

The mixture is drawn directly without evaluating the closed-form EMG density, exploiting the fact that the EMG is the convolution of an independent Gaussian and an exponential:

$$\text{EMG}(\mu, \sigma, \tau) \stackrel{d}{=} \mathcal{N}(\mu, \sigma) + \text{Exp}(1/\tau). \quad (11)$$

Algorithm 2: `sample_lambda_obs(ltr, z, interp, rng)`

Knobs: posterior-mean projection coefficients in `interp`

Inputs: λ^{tr} (per halo), $z = z^{\text{ob}}$ (per halo)

Outputs: λ^{ob} (per halo), and the per-halo $(f_{\text{prj}}, \tau, \mu, \sigma)$

```

1  $(f_{\text{prj}}, \tau, \mu, \sigma) \leftarrow \text{projection\_params}(\lambda^{\text{tr}}, z, \text{interp})$ 
2  $g \leftarrow \mu + \sigma \cdot \text{rng.standard\_normal}()$ 
3  $u \leftarrow \text{rng.uniform}()$ 
4  $b \leftarrow \text{rng.exponential}(\text{scale} = 1/\tau)$ 
5 return  $\lambda^{\text{ob}} = g + b \mathbb{I}[u < f_{\text{prj}}]$ 
```

Numerical recipes. In the closed-form sanity histogram the EMG prefactor is computed in log space ($\log[\tau/2] + (\tau/2)(2\mu + \tau\sigma^2 - 2\lambda^{\text{ob}})$) and exponentiated once, to avoid overflow at large τ , and `scipy.special.erfc` is used directly rather than `erfcx` since the argument here is bounded and small. The clamps $f_{\text{prj}} \in [0, 1]$, $\tau, \sigma \geq 10^{-6}$ in `projection_params` guard against pathological coefficient combinations from outlier posterior draws.

Sanity checks (notebook cells 18–20). Two diagnostics: (a) $\sim 2 \times 10^5$ Monte-Carlo draws at fixed (λ^{tr}, z) are histogrammed against the closed-form C19 PDF on a log- y scale at three redshifts $\{0.1, 0.2, 0.3\}$ and three intrinsic richnesses $\{26, 58, 100\}$; (b) the C19 kernel is decomposed into its $(1 - f_{\text{prj}})\mathcal{N}$ Gaussian core and its f_{prj} EMG projection tail at $z = 0.4$ for $\lambda^{\text{tr}} \in \{10, 30, 60, 100\}$, to visualise the EMG tail dominating the high- λ^{ob} side at large λ^{tr} and driving the projection bias on the lensing data vector.

5 Output 1 — $N(\lambda^{\text{ob}}, z^{\text{ob}})$ number counts

The number-counts data vector is a direct histogram of the C19 forward-model λ^{ob} on the observed-frame z^{ob} :

$$N_{ij}^{\text{fwd}} = \sum_h \mathbb{I}[\lambda_h^{\text{ob}} \in [\lambda_i, \lambda_{i+1})] \mathbb{I}[z_h^{\text{ob}} \in [z_j, z_{j+1})], \quad (12)$$

on the default 4×3 grid above. The two binning quantities (λ^{ob} and z^{ob}) come from the forward model rather than from the catalog redMaPPer columns: the abundance side of the data vector lives in the C26 model world, not in the cluster-finder world. This is the correct entry for a C26 likelihood that integrates Eq. 15 $\rightarrow$ Eq. 6 to predict $N(\lambda^{\text{ob}}, z^{\text{ob}})$.

The headline numbers on the notebook’s default grid are

$$N(\lambda^{\text{ob}}, z^{\text{ob}}) = \begin{pmatrix} 28\,903 & 71\,482 & 90\,166 \\ 4\,138 & 9\,419 & 11\,699 \\ 1\,242 & 2\,739 & 3\,120 \\ 482 & 941 & 903 \end{pmatrix},$$

reflecting the steeply falling Buzzard halo mass function and the photo- z kernel pile-up at intermediate redshift.

6 Stacked $\Sigma(R)$ in λ bins: true vs observed

The lensing data vector is built by averaging per-halo $\Sigma_h(R)$ inside each (λ, z) bin,

$$\langle \Sigma(R) | \lambda_i, z_j \rangle = \frac{1}{N_{ij}} \sum_{h \in \text{bin}} \Sigma_h(R), \quad (13)$$

so the stacked profile is shaped by the *mass distribution within the bin*. Because Σ_h is fixed by halo mass via the NFW-like projected profile, comparing two stacks at the same numerical λ but different binning quantities — one binned on the intrinsic λ^{tr} , one on the observed λ^{ob} — isolates how optical selection alters that mass distribution.

Mass-dilution piece. The λ^{tr} bin samples a narrow mass range set by the C26 mass–richness relation (the “intrinsic” richness–mass relation). The λ^{ob} bin additionally pulls in low-mass halos that got projection-boosted up into the range via the asymmetric C19 EMG tail. The expected ordering is

$$\langle \Sigma | \lambda_i^{\text{ob}} \rangle < \langle \Sigma | \lambda_i^{\text{tr}} \rangle, \quad (14)$$

because the λ^{ob} bin’s mean mass — and hence its mean Σ — is dragged down by the extra low-mass halos. This is the mass-dilution / Eddington-bias piece of the optical-selection effect that the forward model captures *without* an explicit $\mathcal{B}_{\text{sel}}$ correction. The notebook displays the ratio $\langle \Sigma | \lambda^{\text{ob}} \rangle / \langle \Sigma | \lambda^{\text{tr}} \rangle$ in cell 25, which sits below unity at all R in the lower-richness bins.

C26 Fig. 3 cross-check. Cell 27 isolates the $(\lambda^{\text{ob}} \in [15, 25], z^{\text{ob}} \in [0.37, 0.51])$ bin (closest to $(\lambda^{\text{ob}} = 20, z = 0.5)$) and replots $\langle \Sigma(R) \rangle$ in comoving Mpc/ h on the x -axis and $M_{\odot}/(\text{Mpc}/h)^2$ on the y -axis to match the C26 Fig. 3 axes. Visual amplitude and slope agree with the published reference.

7 Halo mass function cross-check

A sanity check on the abundance side of the data vector: the binned $dn/d\log_{10} M$ from the Buzzard catalogue inside the volume

$$V_{\text{sample}} = f_{\text{sky}} \left[V_{\text{com}}(z_{\text{max}}) - V_{\text{com}}(z_{\text{min}}) - V_{\text{com}}(0.37) + V_{\text{com}}(0.33) \right], \quad (15)$$

with $f_{\text{sky}} = 4143/41253$ (DES Y3 Buzzard area), is overlaid against an analytical Tinker 2008 HMF evaluated at the Buzzard cosmology with $\sigma_8 = 0.82$, $n_s = 0.96$, the Eisenstein–Hu transfer function, and the Bryan & Norman 1998 spherical-overdensity virial mass definition (`SOVirial` in `hmf`, matching the Buzzard `Mvir`). The two agree at the 10–20% level over the relevant mass range, which suffices for a likelihood that is dominated by the lensing leg — a future cosmology run will not be pulled away from the input by an HMF mismatch.

8 Output 2 — $\langle \gamma_t(R) \rangle$ and the empirical $\mathcal{B}_{\text{sel}}$

The cluster-finder route to the lensing data vector. The forward model of §4 produces a λ^{ob} that is statistical in (M, z) alone — it carries no per-halo line-of-sight (LSS) information. Optical-selection bias on the lensing profile only appears when the binning quantity itself responds to each cluster’s environment. The Buzzard catalogue already has that quantity: `LAMBDA_CHISQ`, the redMaPPer-measured richness produced by running the cluster finder on the Buzzard photometric mock, and its self-consistent cluster photo- z `Z_LAMBDA`. We use those two columns as the lensing-side bin definition.

8.1 Lensing geometry

The shear-side conversion from $\Delta\Sigma$ to γ_t uses the precomputed β_{eff} table loaded from `floc.mock_boost_factor_1d`:

$$\Sigma_{\text{crit}}^{-1}(z_l) = \frac{4\pi G}{c^2} D_{\text{com}}(z_l) \beta_{\text{eff}}(z_l), \quad (16)$$

with $D_{\text{com}}(z_l)$ from `cosmo0.comoving_distance` and the prefactor $4\pi G/c^2 = 6.01 \times 10^{-19} \text{ pc}/M_{\odot} \times 10^6$ for the pc-to-Mpc conversion that puts the result in $\text{pc}^2/M_{\odot}$. The data vector is then

$$\langle \gamma_t(R) | \lambda^{\text{RM}}, z^{\text{RM}} \rangle = \langle \Delta\Sigma(R) \rangle_{\text{bin}} \Sigma_{\text{crit}}^{-1}(\langle z^{\text{RM}} \rangle_{\text{bin}}). \quad (17)$$

Because the per-halo `DeltaSigma` column was already particle-pair-counted on the Buzzard simulation by the upstream profile pipeline, no $\Sigma \rightarrow \Delta\Sigma$ inversion is performed here.

8.2 The (M, z_{true}) -matched random reference

To isolate the *LSS-correlated* piece of the optical-selection bias from the mass-dilution piece, we follow Wu+2022 / Esteves (`stacked_profile_weighted_by_mass_redshift.py`) and build a random reference whose $(\log_{10} M, z^{\text{tr}})$ distribution matches the cluster-finder-selected sample’s, but is drawn *blind* to the cluster finder. For each $(\lambda^{\text{RM}}, z^{\text{RM}})$ bin and a fixed $(\Delta \log M, \Delta z) = (0.1, 0.05)$ grid:

$$\langle \Delta\Sigma(R) \rangle_{\text{rnd}} = \frac{\sum_c w_c \langle \Delta\Sigma(R) | c \rangle_{\text{full mock}}}{\sum_c w_c}, \quad w_c = N_c^{\text{sel}}. \quad (18)$$

The cell-average is taken over the *full* mock, so the reference resamples halos that share the selected sample’s mass and redshift but were never preferentially picked for their environment. The empirical selection bias is the per-bin ratio

$$\mathcal{B}_{\text{sel}}^{\text{emp}}(R | \lambda^{\text{RM}}, z^{\text{RM}}) = \frac{\langle \Delta\Sigma(R) \rangle_{\text{sel}}}{\langle \Delta\Sigma(R) \rangle_{\text{rnd}}}. \quad (19)$$

$\mathcal{B}_{\text{sel}}^{\text{emp}}$ is the catalog-direct analogue of the C26 Eq. C1 fitting function: no $R \rightarrow 0$ extrapolation, no Σ -vs- $\Delta\Sigma$ calibration assumption, both numerator and denominator come from the same per-halo `DeltaSigma` column.

8.3 Algorithm

Algorithm 3: `lensing_data_vector`(per bin $(\lambda_i^{\text{RM}}, z_j^{\text{RM}})$)

Knobs: $\Delta \log M = 0.1$, $\Delta z = 0.05$, `LDBBINS`, `ZMIN_LIST`, `ZMAX_LIST`

Inputs: `LAMBDA_CHISQ`, `Z_LAMBDA`, `DeltaSigma`, `Mvir`, `redshift_true`, β_{eff} table

Outputs: `shear_data_vector` (selected), `shear_random` (mass-matched), `B_sel_emp`

```

1 for each  $(\lambda_i^{\text{RM}}, z_j^{\text{RM}})$  bin do
2    $m \leftarrow$  halos with LAMBDA_CHISQ in  $[\lambda_i, \lambda_{i+1})$  and Z_LAMBDA in  $[z_j, z_{j+1})$ 
3    $\langle \Delta \Sigma \rangle_{\text{sel}} \leftarrow \text{mean}(\text{DeltaSigma}[m], \text{axis}=0)$ 
4    $\langle \Delta \Sigma \rangle_{\text{rnd}} \leftarrow \text{mass\_match\_stack}(\log_{10} M_{\text{vir}}[m], z^{\text{tr}}[m], \text{DeltaSigma}[m],$ 
       $\log_{10} M_{\text{vir}}, z^{\text{tr}}, \text{DeltaSigma}, \Delta \log M, \Delta z)$ 
5    $\langle z^{\text{RM}} \rangle \leftarrow \text{mean}(\text{Z\_LAMBDA}[m])$ 
6    $\Sigma_{\text{crit}}^{-1} \leftarrow \text{interp}(\langle z^{\text{RM}} \rangle, z_{\text{lens}}, \beta_{\text{eff}} \text{ table})$ 
7    $\text{shear\_data\_vector}_{ij} \leftarrow \langle \Delta \Sigma \rangle_{\text{sel}} \cdot \Sigma_{\text{crit}}^{-1}$ 
8    $\text{shear\_random}_{ij} \leftarrow \langle \Delta \Sigma \rangle_{\text{rnd}} \cdot \Sigma_{\text{crit}}^{-1}$ 
9    $\text{B\_sel\_emp}_{ij} \leftarrow \langle \Delta \Sigma \rangle_{\text{sel}} / \langle \Delta \Sigma \rangle_{\text{rnd}}$ 

```

Per-bin counts. On the default $\{5, 15, 25, 40, 160\} \times \{[0.20, 0.32), [0.37, 0.51), [0.51, 0.64)\}$ grid:

$$N_{\text{sel}} = \begin{pmatrix} 31\,137 & 74\,942 & 80\,092 \\ 2\,096 & 6\,806 & 6\,272 \\ 397 & 1\,367 & 1\,019 \\ 97 & 381 & 235 \end{pmatrix}.$$

The high-richness, high-redshift bins are the limiting factor for the bootstrap precision of $\mathcal{B}_{\text{sel}}^{\text{emp}}$ (cell 44 below).

8.4 Wu+2022 Fig. 2 reproduction

The notebook reproduces the Wu+2022 (arXiv:2203.05416) Fig. 2 panel grid in cells 43–44: per $(\lambda^{\text{RM}}, z^{\text{RM}})$ bin, the selected/ (M, z) -matched ratio is overlaid for both Σ (orange) and $\Delta \Sigma$ (blue) on a log- x axis spanning ~ 0.1 –30 Mpc. The bell-shaped peak around $R \sim 1$ Mpc is the LSS-environment-driven selection bias and reproduces the Wu result.

A 50-sample bootstrap (cell 44) resamples halos within each bin with replacement and re-builds the mass-matched reference at every step to capture the statistical uncertainty driven by finite halo count; in the lowest- λ bins this is the dominant source of the apparent dip and bump structure beneath the bell.

8.5 Null test on the C19 sampler

Cell 47 repeats the same panel grid but bins on the C19 forward-model $(\lambda^{\text{ob}}, z^{\text{ob}})$ in place of the catalog $(\lambda^{\text{RM}}, z^{\text{RM}})$. The selected/ (M, z) -matched ratio collapses to ≈ 1 at all R as expected, because the C19 sampler is by construction blind to per-halo LSS environment. This is a clean null test of the mass-matching procedure: any residual deviation from unity is the bias of the matching algorithm itself, not a physical signal.

9 The analytical Costanzi+2026 Eq. C1 closure

A purely analytical alternative to §8’s catalog-empirical ratio. The C26 Appendix C fitting function for the bias on Σ is

$$\mathcal{B}_{\text{sel},\text{C1}}^{\Delta\Sigma}(R|\lambda^{\text{ob}}, z) = A \left(\frac{R}{R_0} \right)^\alpha \left[1 + \left(\frac{R}{R_0} \right)^\gamma \right]^{(\beta-\alpha)/\gamma} + C, \quad (20)$$

with $R_0 = R_\lambda(\lambda^{\text{ob}})(1+z)$ and $R_\lambda = (\lambda^{\text{ob}}/100)^{0.2} h^{-1} \text{Mpc}$ (C26 Appendix B). The same functional form fits the bias on $\Delta\Sigma$ (Matteo, private communication) with a different parameter set. Both flavours are implemented in the notebook:

flavour	A	α	β	γ	C
C1 for $\Delta\Sigma$ (Matteo, priv. comm.)	0.12	4.11	−0.18	1.82	1.00
C1 for Σ (C26 published)	0.10	0.92	−0.53	4.10	1.00

(21)

The C26 paper writes the Σ form with a fixed +1 rather than a free C , which is the $C = 1$ limit of Eq. (20).

The data vector is then

$$\Delta\Sigma_{\text{obs}}(R) = \mathcal{B}_{\text{sel},\text{C1}}^{\Delta\Sigma}(R|\lambda^{\text{ob}}, z) \Delta\Sigma_{\text{true}}(R), \quad \langle\gamma_t(R)\rangle = \Delta\Sigma_{\text{obs}}(R) \Sigma_{\text{crit}}^{-1}(\langle z^{\text{ob}} \rangle_{\text{bin}}). \quad (22)$$

The $\Delta\Sigma_{\text{true}}$ stack is taken on the C19 forward-model ($\lambda^{\text{ob}}, z^{\text{ob}}$), so the chain $\lambda^{\text{tr}} \rightarrow \lambda^{\text{ob}} \rightarrow \Delta\Sigma_{\text{obs}}$ lives entirely in the analytical-model world — distinct from §8 which stacks on catalog redMaPPer outputs.

Why both routes coexist. The full C26 forward model (Σ_{1h} via Eq. 12 + Σ_{prj} via Eq. 13, then Wright–Brainerd to $\Delta\Sigma$) involves nested line-of-sight integrals through the optical-bias kernel $b_{\lambda^{\text{ob}}}$ (C26 Eq. 8) that are prohibitive at every likelihood evaluation. Eq. (20) collapses that calculation into five constants per richness/redshift bin, which is what makes a production MCMC tractable. The empirical $\mathcal{B}_{\text{sel}}^{\text{emp}}$ of §8 is the cross-check that the analytical form actually fits the simulation: the two should agree on a per-bin basis to within the bootstrap uncertainty of the empirical construction.

Matteo’s binning preset. Cell 50 re-runs the analytical C1 evaluation on the alternative DES Y3-style grid `LBDBINS_M = {20, 30, 45, 60, 500}`, `ZMIN_LIST_M = {0.20, 0.35, 0.50}`, `ZMAX_LIST_M = {0.35, 0.50, 0.65}`, and plots the $\mathcal{B}_{\text{sel},\text{C1}}^{\Delta\Sigma}$ and $\mathcal{B}_{\text{sel},\text{C1}}^\Sigma$ panel grid on the comoving-Mpc/ h axis with a y -range fixed to $[0.97, 1.13]$ for direct comparison to the C26 reference figure.

10 Output archive

Cell 52 writes the entire data vector to a single `.npz` archive at `/global/u2/x/xintang/mocks/mock/Xin_MockDataVector`

Key	Shape	Content
<code>radii_phys_mpc</code>	$(N_R,)$	15-bin physical radial grid (Mpc)
<code>lambda_bins</code>	$(N_\lambda + 1,)$	richness edges $\{5, 15, 25, 40, 160\}$
<code>z_bin_min</code>	$(N_z,)$	lower-redshift edges
<code>z_bin_max</code>	$(N_z,)$	upper-redshift edges
<code>number_counts</code>	(N_λ, N_z)	C19 forward-model $N(\lambda^{\text{ob}}, z^{\text{ob}})$
<code>counts_per_bin</code>	(N_λ, N_z)	redMaPPer $N(\lambda^{\text{RM}}, z^{\text{RM}})$
<i>Catalog redMaPPer + mass-matched ratio path (§8)</i>		
<code>DSigma_selected</code>	(N_λ, N_z, N_R)	$\langle \Delta \Sigma \rangle_{\text{sel}}$ on $(\lambda^{\text{RM}}, z^{\text{RM}})$
<code>DSigma_random</code>	(N_λ, N_z, N_R)	(M, z^{tr}) -matched random reference
<code>B_sel_emp</code>	(N_λ, N_z, N_R)	empirical $\mathcal{B}_{\text{sel}}^{\text{emp}}$ ratio
<code>shear_data_vector</code>	(N_λ, N_z, N_R)	$\langle \gamma_t \rangle_{\text{sel}}$
<code>shear_random</code>	(N_λ, N_z, N_R)	$\langle \gamma_t \rangle_{\text{rnd}}$
<i>C19 forward-model + analytical C1 path (§9)</i>		
<code>DSigma_lob_C19</code>	(N_λ, N_z, N_R)	$\langle \Delta \Sigma \rangle$ on C19 $(\lambda^{\text{ob}}, z^{\text{ob}})$
<code>Bsel_DSigma_C1</code>	(N_λ, N_z, N_R)	C26 Eq. C1 fit on $\Delta \Sigma$
<code>Bsel_Sigma_C1</code>	(N_λ, N_z, N_R)	C26 Eq. C1 fit on Σ (overlay)
<code>DSigma_obs_C1</code>	(N_λ, N_z, N_R)	$\Delta \Sigma_{\text{lob}}^{\text{C19}} \cdot \mathcal{B}_{\text{sel,C1}}^{\Delta \Sigma}$
<code>shear_C19_C1</code>	(N_λ, N_z, N_R)	analytical observed shear
<code>shear_C19_stack</code>	(N_λ, N_z, N_R)	no-correction reference (C19 stack only)

Table 2: Data-vector archive layout. Default shape is $N_\lambda = 4$, $N_z = 3$, $N_R = 15$ (60-element scalar vectors, 180-element radial vectors).

11 Known simplifications

- **Photo- z kernel.** The Gaussian $\sigma_z = 0.01 (1 + z^{\text{tr}})$ proxy is a stand-in for the redshift- and colour-dependent kernel of Myles+2021. It captures the dominant scatter but misses the tails that drive the high- z outlier rate.
- **Boost-factor contamination.** Cluster-member contamination of the source sample (Y3 boost factors) is not yet modelled. This is the other major Y3 lensing systematic that needs to be added on top of the optical-selection bias.
- **Posterior-mean projection.** The C19 projection parameters are evaluated at the posterior mean of the upstream chain. Row-by-row covariance propagation (which would convert the parameter posterior into a per-bin λ^{ob} distribution) is not implemented.
- **Masking and miscentering.** The masking fraction f_{msk} from C19 is loaded but unused; centering mis-identification (which is the focus of §9 of Esteves’ `pipeline_modules.tex`) is not modelled here.
- **Single-scale $\mathcal{B}_{\text{sel}}$ in `costanzi_selection.py`.** The legacy `b_sel_one_halo` weight in the helper module is the small-scale (one-halo) limit of C26 Appendix C; it is superseded in the production notebook by the scale-dependent Eq. (20) fitting function, but left in the helper for completeness.

12 File map

- `Xin.MockDataVector.ipynb` — the production notebook. Top-to-bottom execution reproduces every figure and saves the data vector.

- `costanzi_selection.py` — C26 Eq. 15 sampler and C19 Eq. 6 mixture sampler, plus the legacy single-scale $\mathcal{B}_{\text{sel}}$ weight. All hard-coded constants are the DES Y1 best-fit values listed in §3.
- `stacked_profile_weighted_by_mass_redshift.py` — Wu/Esteves (M, z)-matched random reference builder.
- `radial_bins_phys_mpc.py` — the 15-bin physical radial grid `rp_phys_mpc`.
- `prj_params_DESY3_lss_lin_dep_getdist_v1.txt` — 15 posterior samples of the 10 C19 projection coefficients.
- `fileLoc.py` — machine-resolved path table for the raw Buzzard halo catalogue, profile output, and β_{eff} table.
- `XinMockDataVector.tex` — this document.

13 References

- Costanzi et al. 2026, *Forward analytical model for the optical selection bias on galaxy cluster lensing profiles*. Eqs. 15 (mass–richness), Appendix C Eq. C1 (selection-bias fit), Sec. III (Poisson–Gaussian convolution).
- Costanzi et al. 2019, arXiv:1807.07072, *Modeling projection effects in optically-selected cluster catalogues*. Eq. 6 (Gaussian + EMG mixture), Appendix A Eq. A8 (per-parameter forms).
- Wu et al. 2022, arXiv:2203.05416. Fig. 2 (selection-bias panel grid) and Appendix B (mass-redshift weighted reference).
- Esteves’ `pipeline_modules.tex` — companion technical reference for the C++/CosmoSIS production pipeline that consumes the data vector saved here.